Caesar Cipher, Encryption

A=0, B=1, C=2, D=3, E=4, F=5, G=6, H=7, I=8, J=9
K=10, L=11, M=12, N=13, O=14, P=15, Q=16, R=17, S=18, T=19
U=20, V=21, W=22, X=23, Y=24, Z=25

- Static shifting: $y_i = (x_i - 5) \bmod 26$

plaintext: LEGEND

Model: $y_i = (x_i - 5) \bmod 26$

then $i=0$ $y_0 = x_0 - 5 = L - 5$
$= 11 - 5 = 6$
$= G$

$i=1$ $y_1 = x_1 - 5 = E - 5$
$= 4 - 5 = -1$
so, $-1 + 26 = 25 = Z$

$i=2$ $y_2 = x_2 - 5 = G - 5$
$= 6 - 5 = 1$
$= B$

$i=3$ $y_3 = x_3 - 5 = E - 5$
$= 4 - 5 = -1$
so, $-1 + 26 = 25 = Z$

$i=4$ $y_4 = x_4 - 5 = N - 5$
$= 13 - 5 = 8$
$= I$

$i=5$ $y_5 = x_5 - 5 = D - 5$
$= 3 - 5 = -2$, so $-2 + 26 = 24$
$= Y$

Therefore: ciphertext = G2B2Y

- Dynamic Shifting: $y_i = (3x_i + 6 - 5i^2) \bmod 26$

plaintext: LEGEND

Model: $y_i = (3x_i + 6 - 5i^2) \bmod 26$

$i=0$, $x_0 = L = 11$ $\Rightarrow y_0 = (3(11) + 6 - 5(0)^2) \bmod 26$
$= (33 + 6 - 0) \bmod 26$
$= 39 \bmod 26 = 13$
$= N$

$i=1$, $x_1 = E = 4$ $\Rightarrow y_1 = (3(4) + 6 - 5(1)^2) \bmod 26$
$= (12 + 6 - 5) \bmod 26$
$= 13 \bmod 26 = 13$
$= N$

$i=2$, $x_2 = G = 6$ $\Rightarrow y_2 = (3(6) + 6 - 5(2)^2) \bmod 26$
$= (18 + 6 - 20) \bmod 26$
$= 4 = E$

$i=3$, $x_3 = E = 4$ $\Rightarrow y_3 = (3(4) + 6 - 5(3)^2) \bmod 26$
$= (12 + 6 - 45) \bmod 26$
$= -27 \bmod 26 = -1$
So, $-1 + 26 = 25 = Z$

$i=4$, $x_4 = N = 13$ $\Rightarrow y_4 = (3(13) + 6 - 5(4)^2) \bmod 26$
$= (39 + 6 - 80) \bmod 26$
$= -35 \bmod 26 = -9$
So, $-9 + 26 = 17 = R$

$i=5$, $x_5 = D = 3$ $\Rightarrow y_5 = (3(3) + 6 - 5(5)^2) \bmod 26$
$= (9 + 6 - 125) \bmod 26$
$= -110 \bmod 26 = -6$
So, $-6 + 26 = 20 = U$

Therefore = NNE2RU

+ Decryption

- Static shifting

decryption formula: $x_i = (y_i + 5) \bmod 26$

ciphertext: G2B2IY

We have

G: $x_0 = (6 + 5) \bmod 26 = 11 = L$

2: $x_1 = (25 + 5) \bmod 26 = 4 = E$

B: $x_2 = (1 + 5) \bmod 26 = 6 = G$

2: $x_3 = (25 + 5) \bmod 26 = 4 = E$

I: $x_4 = (8 + 5) \bmod 26 = 13 = N$

Y: $x_5 = (24 + 5) \bmod 26 = 3 = D$

Therefore: G2B2IY → LEGEND

- Dynamic Shifting

decryption formula: $x_i = 9(y_i - 6 + 5i^2) \bmod 26$

ciphertext: NNE2RU

$i=0, y_0 = N = 13$ $\Rightarrow x_0 = 9(13 - 6 + 5(0)^2) \bmod 26$
$= 63 \bmod 26 = 11 = L$

$i=1, y_1 = N = 13$ $\Rightarrow x_1 = 9(13 - 6 + 5(1)^2) \bmod 26$
$= 108 \bmod 26 = 4 = E$

$i=2, y_2 = 6 = 4$ $\Rightarrow x_2 = 9(4 - 6 + 5(2)^2) \bmod 26$
$= 162 \bmod 26 = 6 = G$

$i=3, y_3 = 2 = 25$ $\Rightarrow x_3 = 9(25 - 6 + 5(3)^2) \bmod 26$
$= 576 \bmod 26 = 4 = E$

$i=4, y_4 = R = 17$ $\Rightarrow x_4 = 9(17 - 6 + 5(4)^2) \bmod 26$
$= 819 \bmod 26 = 13 = N$

$i=5, y_5 = U = 20$ $\Rightarrow x_5 = 9(20 - 6 + 5(5)^2) \bmod 26$
$= 1251 \bmod 26 = 3 = D$

Therefore NNE2RU → LEGEND